with two different modes of benchmarking. One is called Benchmark and other Timedemo mode.

The Benchmark mode delivers figures that are not related to the game at all. Basically, the test examines your video card and returns actual values for details such as fillrate and geometry speed. With Timedemo, the game plays back a sequence of in-game frames and calculates how quickly the video card can render and display them onscreen.

To begin your evaluation with *Serious Sam* in Benchmark mode, enter console mode by hitting the [Esc] key once. Then type ***/benchmark()*** and hit the [Enter] key. For the Timedemo, bring down the console in the main menu and type ***/dem_bprofile 1***, hit the [Enter] key and select a demo you wish to run. Once the loop is completed, revert back to console again to find your score.

What to expect

500 MHz CPU with GeForce2: 44 fps
1 GHz CPU with GeForce2: 60 fps
1.5 GHz CPU with GeForce2: 74 fps
(Tested at 1024x768 with 32-bit colour)

Warning

Ninety per cent of the bugs reported with *Serious Sam* are due to bugs of OpenGL drivers. If you have a card that's over a year old, it would be wise to get your drivers up to speed so that the game actually works!

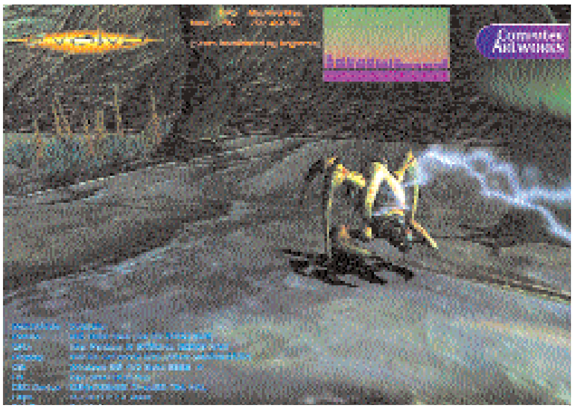
System Prerequisites

AMD K6-3 400 MHz, or 300 MHz Pentium II chip, 64 MB RAM, OpenGL or DirectX 8 3D accelerator with 16 MB, 200 MB free disk space

Evolva

Evolva is a game that uses Microsoft'sDirect 3D API and, like *Serious Sam*, the title heavily checks the T&L performance of your video card. *Evolva* supports a wide range of DirectX 7 options including large textures, bump mapping and hardware T&L. It has also proved itself to be well suited to newer chips, both from AMD and Intel, since the underlying code has been heavily optimised for SSE multimedia instructions.

There are two *Evolva* benchmarks available: a standalone Rolling demo which also features a benchmarking mode, and a patch for the retail version of *Evolva*. We will be using the Rolling demo for benchmarking. After you've installed *Evolva*, navigate to its program group through the Start menu. Click and execute the *Evolva* Rolling demo with bump mapping. The test infinitely loops, so once one segment is complete and while it attempts to load again, press the [Esc] key—this will break the demo and show a text file filled with results.



Evolva Rolling Demo is perfectly suited to test the older generation of graphics techonology

What to expect

500 MHz CPU with GeForce2: 70 fps
1 GHz CPU with GeForce2: 94 fps
1.5 GHz CPU with GeForce2: 115 fps
(Tested at 1024x768 with 32-bit colour.)

Warning

With the Rolling demo comes two separate tests. One test uses bump mapping while the other one doesn't. Aside from this detail, both the tests are essentially the same. If you have a graphics cards based upon the GeForce2 (or higher) or ATi's Radeon, opt for the choice with bump mapping; owners of the original GeForce 256 and S3's Savage 2000 should use the one without it.

System Prerequisites

233 MHz processor, 32 MB RAM, 200 MB free hard disk space, DirectX 7 and a video card with 16 MB RAM

3DMark 2001SE

MadOnion's 3DMark 2001 SE includes many tests monitoring several aspects of your graphics card. From hardware transforms to polygon-crunching large scenes, 3DMark 2001 SE really works with 3D hardware to the fullest. Once you've installed the software and are ready to begin testing, the first thing you need to do is to disable all running programs. Close everything in the system tray. Also, keep in mind the more invisible tasks running in the background that might not be so obvious to locate. In order to get rid of these, hit the [Ctrl] + [Alt] + [Delete] keys which brings up Windows Task Manager. From here, you can disable most unnecessary processes that are running in the background.

Once you have that out of the way, it's time to start benchmarking your graphics card. Fire up 3DMark 2001 SE. Now you have to set up and specify which tests to run. Click the 'Tests' button from the main menu and check all the areas that apply. Unregistered users of the 3DMark software can still carry out all performance-related tests, minus the image quality tests, which hold no bearing on 3D performance in the first place.

Loop through the series of tests for at least three times, where your final score would be the average of the three test runs. If your system crashes before the testing is done, reboot your computer and re-run the test from the beginning after carrying out the necessary steps for getting an optimum score.

To see where you stand, 3DMark 2001 SE includes an online result browser that lets you compare your score with other users that have a similar hardware configuration.

What to expect

Naturally, you'll be looking for a comparative figure to hold up against your own results. When your scores fall between different system configurations, your score should be somewhat proportionate to your PC's place in the table.
500 MHz CPU with GeForce3: 3800
1 GHz CPU with GeForce3: 4960
1.5 GHz CPU with GeForce3: 6182
(Tested at 1024x768 with 32-bit colour)

Warning

If you want 3DMark to complete its tests successfully, ensure that the test settings are set to something more down to earth.